

Using Data to Verify Whether Ann Taylor Clothes Are True to Size for Women in Washington State

CSE 163 Final Project Report | Elly Lee | Student #2163622

1. Summary of Questions and Results

The following is a brief summary of each research question and its answer. Detailed results and justifications are provided in Section 7.

RQ	Question	Answer
RQ 1	How do the average body measurements of Washington state women compare to the U.S. national average?	Washington state women have a slightly smaller waist (~9.9mm) and lower weight (~1.5kg) than the national average, but structurally identical sleeve length, inseam, and shoulder measurements.
RQ 2	How do Washington state women's measurements align with Ann Taylor's published size guide, and where are the greatest mismatches?	Over 50% of WA women — and U.S. women generally — exceed Ann Taylor's published size range by waist measurement. WA women are marginally better served than the national average, but the difference is negligible (<1 pp per size).
RQ 3	Is Washington state's demographic composition distinct enough from the U.S. average to warrant region-specific sizing considerations?	Yes. WA has notably more Asian women (10.3% vs. 6.1% nationally) and fewer Black (3.5% vs. 12.3%) and Hispanic (14.0% vs. 18.9%) women, making its demographic profile meaningfully distinct from the U.S. national average.
RQ 4	Are the body measurement differences between WA women and the U.S. average statistically significant enough to justify region-specific sizing adjustments?	Waist and weight differences are statistically significant ($p < 0.05$), but all Cohen's d effect sizes are negligible ($d < 0.1$). This does not justify region-specific sizing adjustments based on body measurements alone.

2. Motivation

The U.S. women's apparel industry is largely calibrated to the body proportions of white American women. Washington state has one of the highest concentrations of Asian-American residents of any U.S. state — a population whose women tend to have smaller and shorter body proportions on average. However, women of other ethnicities in Washington may also be more petite, and this project does not assume Asian-American women are the sole driver of any observed difference.

As a sales associate at the Ann Taylor location in Bellevue, WA — the only Ann Taylor store in the entire state — I have directly observed that "too big/small" is consistently the most frequently selected return reason in our store's return processing system. Because this single store serves all of Washington state, a sizing mismatch here affects Ann Taylor's entire WA customer base. Notably, Ann Taylor already carries a Petite fit category — yet returns citing size mismatch persist even on petite products, suggesting the existing petite line is still not adequately proportioned for Washington state women. This project aims to quantify that gap with data, and to provide a measurable foundation for more targeted sizing decisions in the Pacific Northwest market.

A secondary motivation emerged during the analysis: Ann Taylor publishes only three to four body measurements per size (bust, waist, hip, and sleeve), compared to Korean retailers such as Musinsa, which routinely publish up to thirteen garment measurements per product — including shoulder width, armhole circumference, thigh width, rise length, and total length. This gap in measurement detail may be a significant driver of returns independent of any regional body size difference.

3. Datasets

Dataset	Source	Size	Used For
ANSUR II (Women), 2012	openlab.psu.edu/ansur2	1,986 rows 99 columns	Structural measurements: sleeve, inseam, shoulder width
NHANES 2015-2018 (Women)	openlab.psu.edu/nhanes	5,449 rows 13 columns	Waist, height, weight by race/ethnicity
ACS PUMS 2020-2024	census.gov/programs-surveys/acs/microdata/access.html	WA: 196,332 rows National: 8.18M rows	Female racial/ethnic demographic weights
Ann Taylor Size Guide (Classic + Petite)	anntaylor.com — via "Size Chart" on any product page	4 DataFrames 10-11 rows each	Reference sizing standard; manually transcribed May 2026

The Ann Taylor size guide was manually transcribed from the size chart modal accessible via any product page at anntaylor.com. It was expanded from the original proposal's single DataFrame into four separate DataFrames — `classic_tops`, `classic_bottoms`, `petite_tops`, `petite_bottoms` — after discovering the Curvy fit subcategory for bottoms and that Classic and Petite bottoms extend to different maximum sizes (size 18 and 18P respectively, while Petite tops max at 16P). All measurements were converted from inches to cm (* 2.54). The national ACS PUMS file was split into four parts (`psam_pusa-d.csv`) requiring concatenation via `pd.concat()`.

4. Method

The analysis was implemented across two Python modules: **analysis.py** (helper functions for loading, cleaning, summarizing, visualizing, and statistical testing) and **eda.py** (main pipeline). The key methodological steps are described below.

Demographic weighting (RQ1, RQ3): ACS PUMS person weights (PWGTP) were used to compute the proportional racial/ethnic composition of Washington state's and the U.S.'s female populations. These weights were applied to NHANES group means (grouped by race_label derived from RIDRETH3) via compute_weighted_means() to produce Washington-representative vs. national average body measurement estimates. Weighted standard deviations were computed via compute_weighted_std() for use in subsequent filtering.

ANSUR II proxy filtering (RQ1): Since ANSUR II contains no race/ethnicity variable, Washington state structural measurements could not be directly weighted. Instead, ANSUR II was filtered to servicewomen whose waist circumference, height, and weight fell within one standard deviation of the Washington-weighted NHANES means, using Washington-weighted standard deviations as the filter range. An equivalent national proxy subgroup was constructed using national-weighted means and standard deviations. These two subgroups were compared across seven structural measurements.

Ann Taylor size alignment (RQ2): Each NHANES respondent was assigned a demographic weight based on their race/ethnicity group's PUMS proportion. Women were then assigned to Ann Taylor size buckets based on waist circumference (converted from mm to cm). An "Above Range" bucket captured women exceeding Ann Taylor's maximum published size. Weighted proportions per size bucket were computed separately for Washington and national profiles.

Statistical significance testing (RQ4): NHANES data was resampled (n=2,000) proportionally to WA and national demographic weights to create two synthetic samples. Independent samples t-tests (scipy.stats.ttest_ind) were run for BMXWAIST, BMXHT, and BMXWT. For ANSUR II structural measurements, t-tests were run directly between the WA and national proxy subgroups. Cohen's d effect sizes were computed alongside p-values using a significance threshold of $\alpha = 0.05$.

5. EDA Summary

The EDA revealed several key findings that shaped the final analysis. ANSUR II contained no missing data across all 99 columns. NHANES had 9,304 missing values, but the majority came from WTMEC4YR (100% missing, unused) and RIDEXPRG (60% missing, irrelevant). Key measurement columns had at most 7% missingness, handled by dropping rows with missing values. Both ACS PUMS files had zero missing values in all key columns.

The most important EDA discovery was that **ANSUR II contains no race/ethnicity variable** — only a Branch column (military job category). This made demographic weighting impossible for structural measurements, requiring the proxy filtering approach described in Section 4. A second key discovery was the **Curvy fit subcategory** in Ann Taylor's bottoms guide, which required expanding the size guide from one DataFrame to four.

The NHANES race/ethnicity breakdown (Figure 1) provided early evidence for the hypothesis: Non-Hispanic Asian women had the smallest mean waist (864.4mm) and weight (61.0kg) of all groups. The PUMS demographic comparison (Figure 2) confirmed Washington state's notably higher Asian female population (10.3% vs. 6.1% nationally), providing a demographic basis for the expected measurement differences.

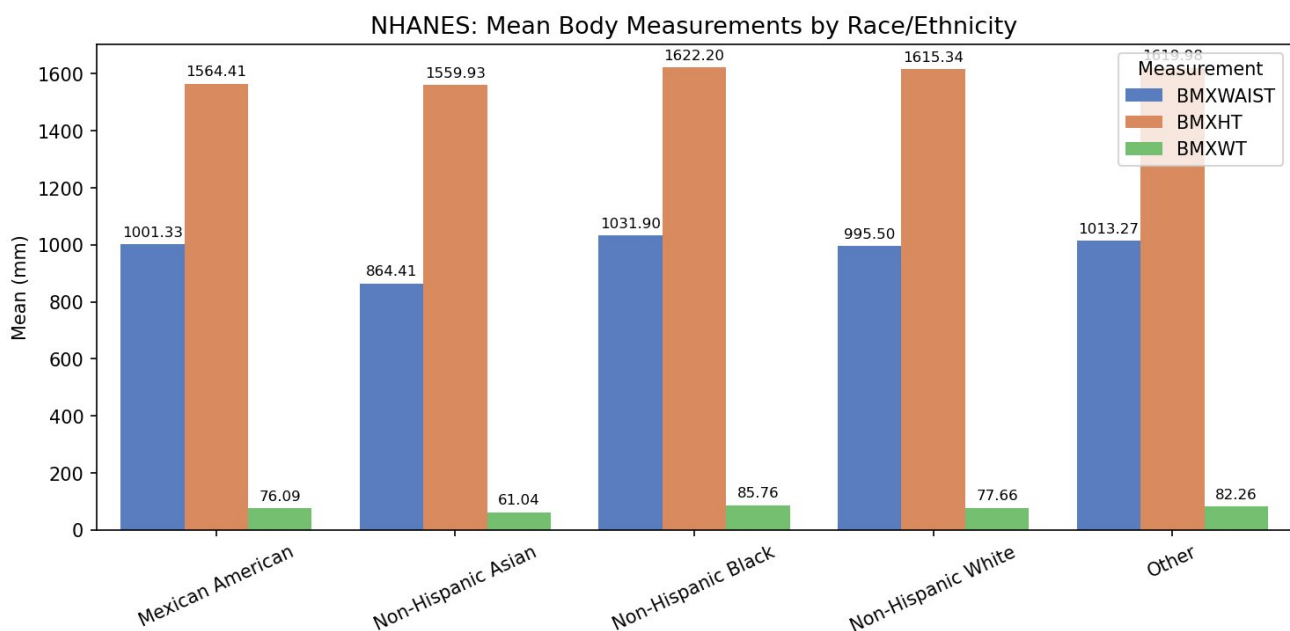


Figure 1. Mean NHANES body measurements by race/ethnicity. Non-Hispanic Asian women show the smallest waist (864.4mm) and weight (61.0kg), motivating the hypothesis that Washington state's higher Asian population contributes to smaller average measurements.

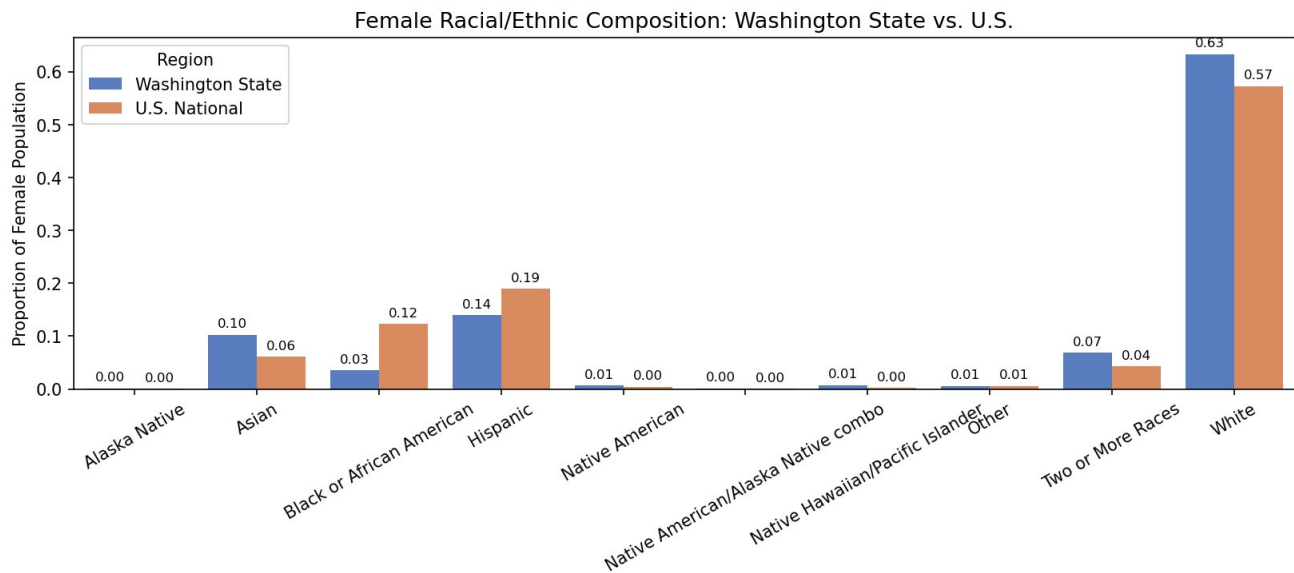


Figure 2. Female racial/ethnic composition of Washington state vs. the U.S. national population, computed from ACS PUMS 2020-2024. Washington state has notably more Asian women (10.3% vs. 6.1%) and substantially fewer Black (3.5% vs. 12.3%) and Hispanic (14.0% vs. 18.9%) women.

6. Results

6.1 RQ1 — Body Measurement Comparison

Applying Washington state's demographic weights to NHANES group means produced the following Washington-representative vs. national average estimates for circumference measurements:

Measurement	WA Weighted Mean	U.S. National Mean	Difference
BMXWAIST (waist, mm)	983.2	993.1	-9.9 mm
BMXHT (height, mm)	1601.7	1602.6	-0.9 mm
BMXWT (weight, kg)	75.9	77.4	-1.5 kg

Washington state women show a consistently smaller waist circumference (~9.9mm) and lower weight (~1.5kg) than the national average, while height is virtually identical. Figure 3 visualizes these differences.

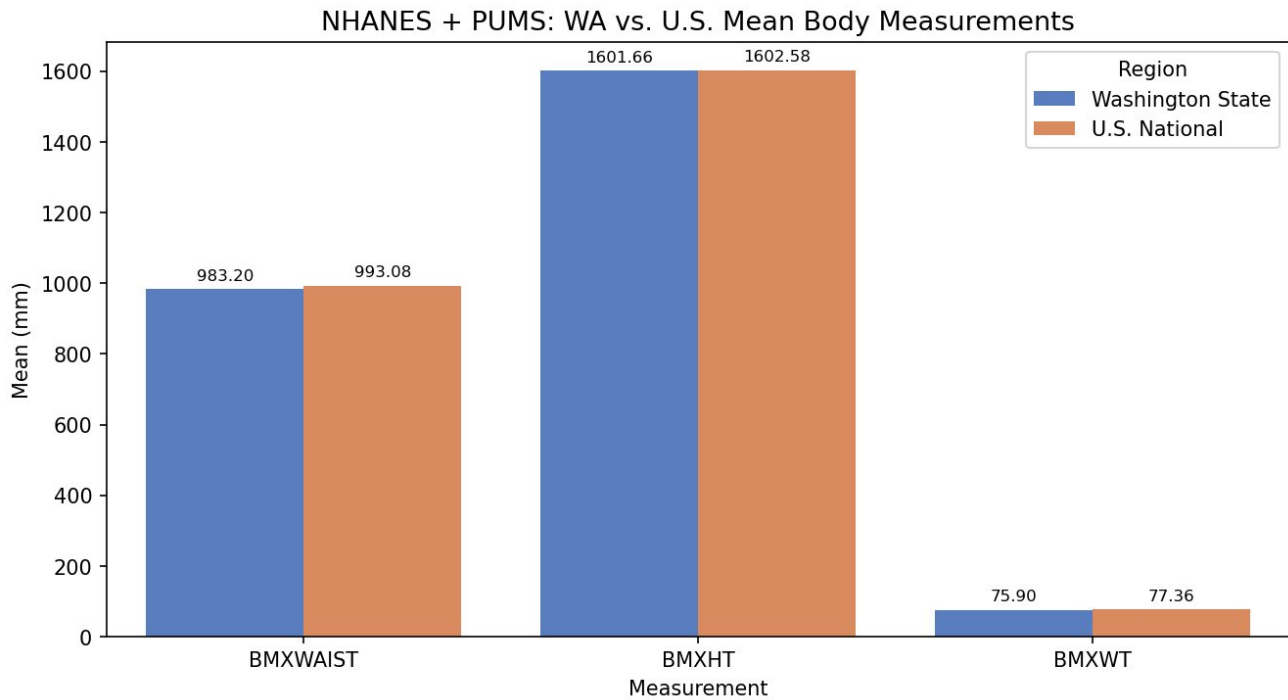


Figure 3. Washington state-weighted vs. U.S. national mean body measurements from NHANES + ACS PUMS. WA women have a smaller estimated waist (983.2 vs. 993.1mm) and lower weight (75.9 vs. 77.4kg), while height is nearly identical.

For structural measurements, the ANSUR II proxy approach filtered 846 WA-representative and 804 national-representative servicewomen. Figure 4 shows the differences across seven structural measurements. All differences are less than 2mm, with thigh circumference showing the largest gap (-1.97mm) and shoulder length the smallest (-0.08mm).

RQ1: WA vs. U.S. Structural Measurement Differences (ANSUR II Proxy)

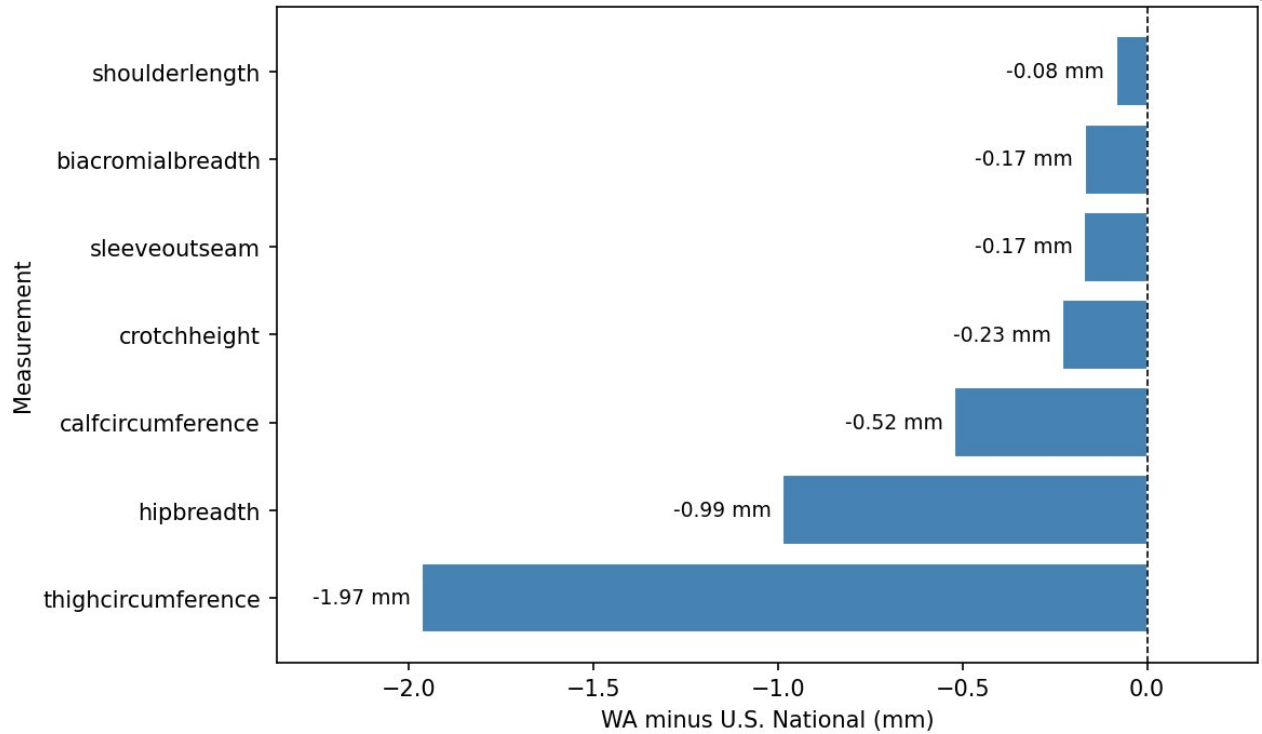


Figure 4. Differences in structural measurements between the Washington state proxy subgroup and national proxy subgroup from ANSUR II. All differences are negative (WA smaller) but less than 2mm across all seven measurements, indicating virtually identical skeletal structure.

Interpretation: Washington state women differ from the national average primarily in body circumference — specifically waist and weight — rather than in skeletal frame dimensions. Sleeve length, inseam, and shoulder width are virtually identical between the two populations when controlling for similar body size profiles. This suggests that any sizing adjustments for Washington state women should focus on waist-related dimensions rather than structural proportions.

6.2 RQ2 — Ann Taylor Size Alignment

Each NHANES respondent was assigned to an Ann Taylor size bucket based on waist circumference, with a separate "Above Range" bucket for women exceeding Ann Taylor's maximum published size. Figures 5 and 6 show the Classic and Petite tops distributions respectively.

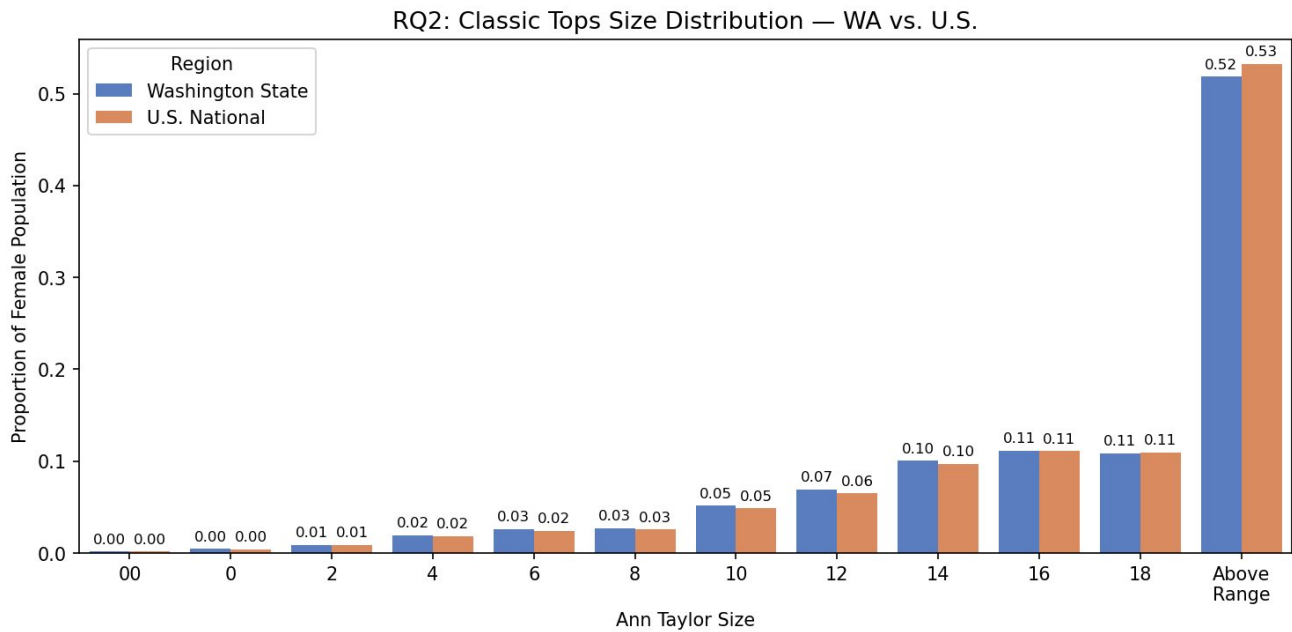


Figure 5. Proportion of Washington state vs. U.S. national women in each Ann Taylor Classic Tops size bucket. 52% of WA women and 53% of U.S. women exceed the Classic size range entirely, with WA women marginally more likely to fall within the published range.

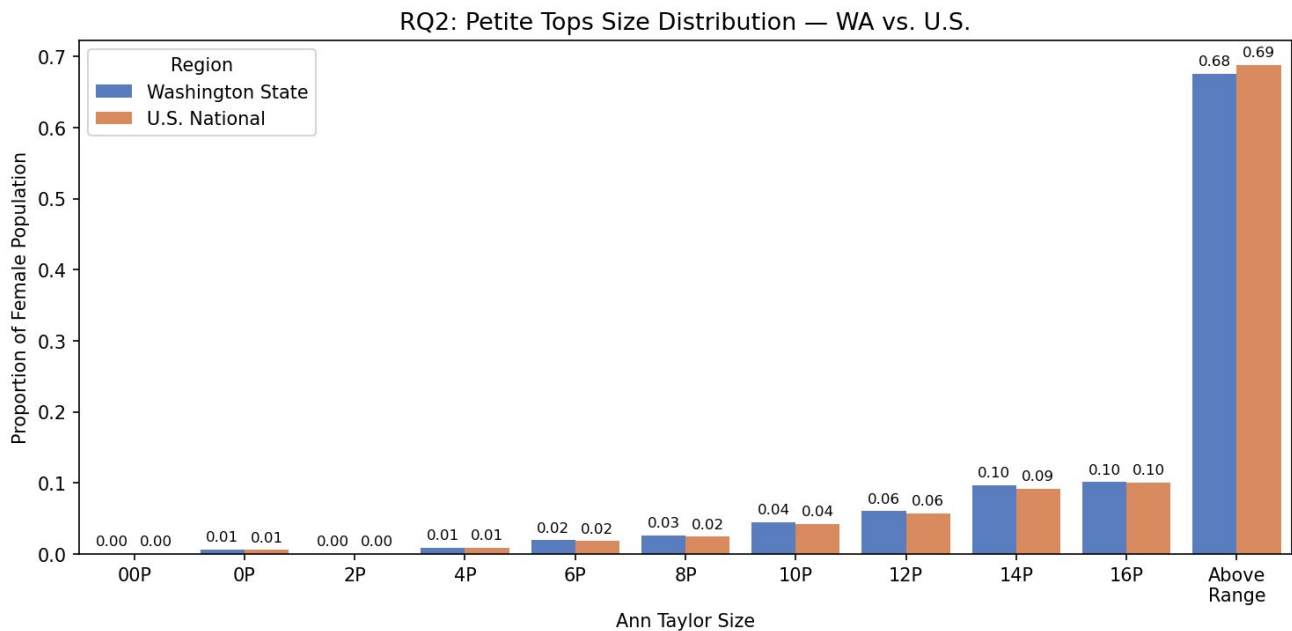


Figure 6. Proportion of Washington state vs. U.S. national women in each Ann Taylor Petite Tops size bucket. 68% of WA women and 69% of U.S. women exceed even the Petite size range, reflecting the narrower waist range of the Petite line (maximum size 16P).

Interpretation: The most striking finding of RQ2 is that the majority of American women — Washington state included — fall completely outside Ann Taylor's published size range by waist measurement. Contrary to the original hypothesis, Washington state women are marginally *more* likely to fall within Ann Taylor's size range than the national average, not less. The WA vs. U.S.

difference at each individual size point is less than one percentage point — negligible for practical purposes. This finding reframes the project's central question: the sizing problem is not a Washington state-specific issue, but a systemic failure of Ann Taylor's published size range to reflect the modern American female population. Furthermore, Ann Taylor's size guide captures only waist, bust, and hip — omitting dimensions such as sleeve length, shoulder width, and thigh circumference that are known drivers of fit dissatisfaction. Korean retailers such as Musinsa routinely publish up to thirteen garment measurements per product, providing customers far more precise fit information.

6.3 RQ3 — Demographic Divergence

Figure 7 shows the divergence between Washington state's and the U.S. national female racial/ethnic proportions, sorted by magnitude. Black or African American shows the largest divergence (8.8 percentage points), followed by White (6.0pp), Hispanic (4.9pp), and Asian (4.2pp).

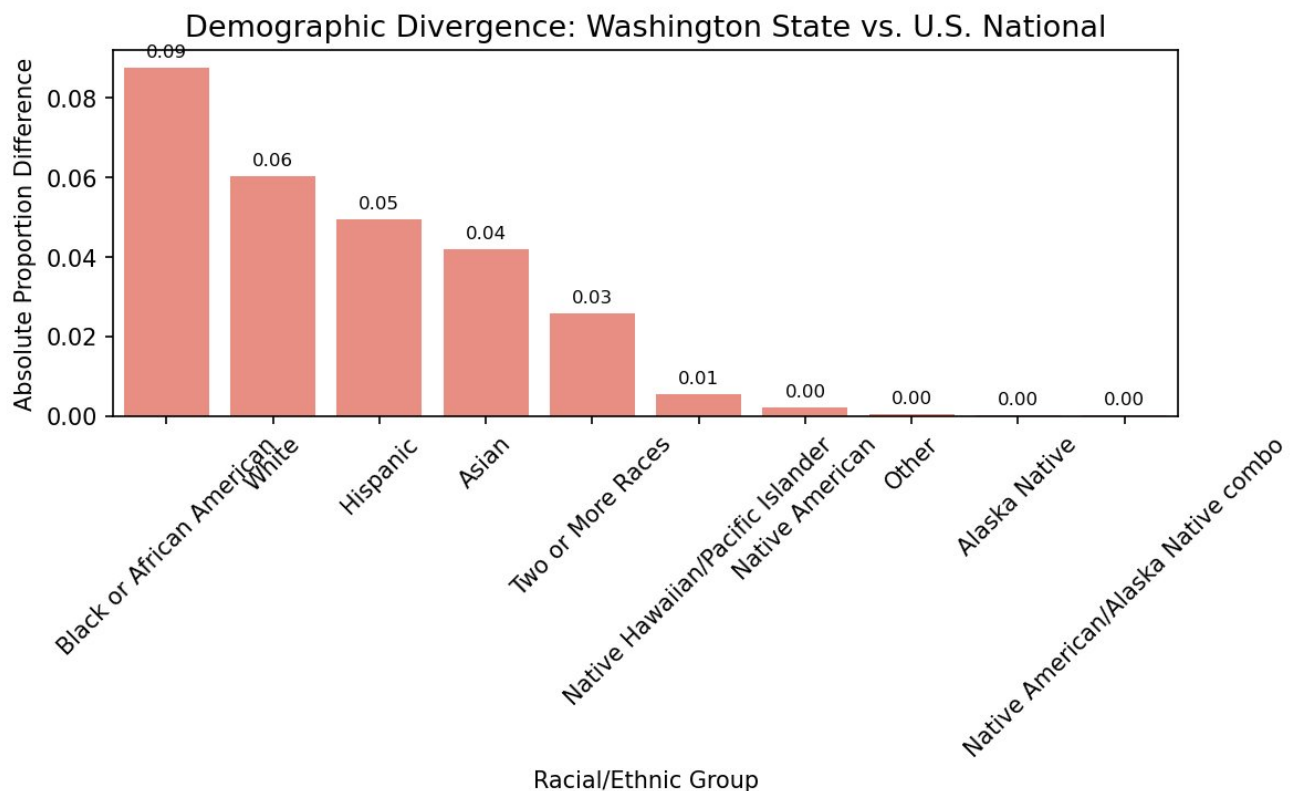


Figure 7. Absolute divergence between Washington state and national female racial/ethnic proportions. Black or African American shows the largest divergence (8.8pp), followed by White (6.0pp), Hispanic (4.9pp), and Asian (4.2pp).

Interpretation: Washington state's female demographic composition is meaningfully distinct from the national average. Notably, the largest divergence is not for Asian women — as the original hypothesis emphasized — but for Black women, of whom Washington has far fewer than the

national average. Since Black women tend to have larger average body measurements than Asian women, Washington's lower Black and higher Asian populations work in the same direction: both contribute to a slightly smaller average waist. This demographic distinctiveness justifies treating Washington state as a meaningfully different market for retail sizing purposes, even if the resulting measurement differences are modest.

6.4 RQ4 — Statistical Significance

Independent samples t-tests were run on NHANES-derived WA and national resampled populations (n=2,000 each), as well as on the ANSUR II proxy subgroups. Figure 8 shows Cohen's d effect sizes for all measurements, colored by statistical significance.

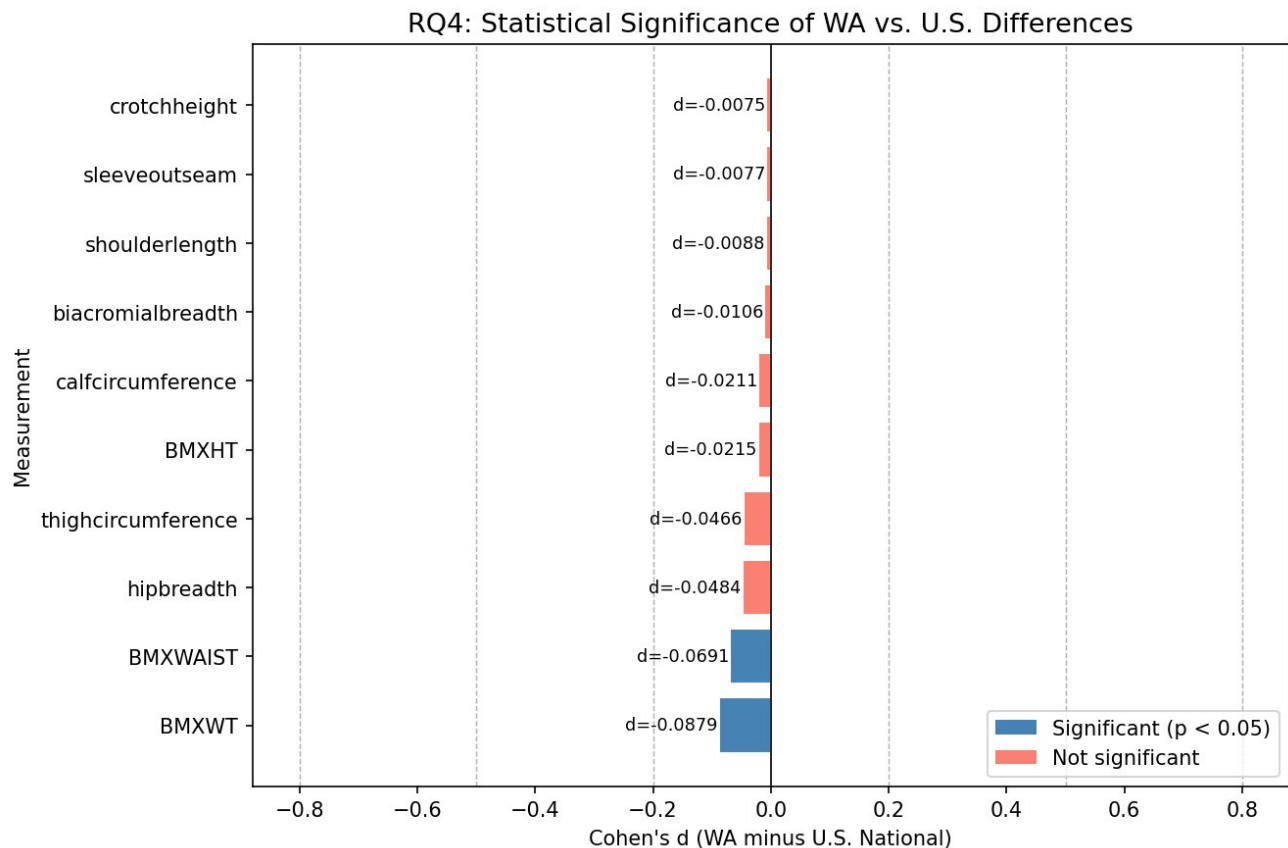


Figure 8. Cohen's d effect sizes for all measurements (WA minus U.S. national). Steelblue bars indicate statistically significant differences ($p < 0.05$); salmon bars indicate non-significant differences. Dashed reference lines mark small ($d=0.2$), medium ($d=0.5$), and large ($d=0.8$) effect benchmarks. All effect sizes are negligible ($d < 0.1$).

Measurement	WA Mean	U.S. Mean	t-stat	p-value	Significant	Cohen's d
BMXWAIST (mm)	984.2	996.3	-2.11	0.035	Yes	-0.069
BMXHT (mm)	1602.4	1603.9	-0.68	0.499	No	-0.022
BMXWT (kg)	75.8	77.7	-2.76	0.006	Yes	-0.088

sleeveoutseam (mm)	538.6	538.7	-0.16	0.876	No	-0.008
crotchheight (mm)	769.8	770.0	-0.15	0.879	No	-0.008
biacromialbreadth (mm)	364.4	364.5	-0.22	0.830	No	-0.011
thighcircumference (mm)	635.0	636.9	-0.95	0.345	No	-0.047
hipbreadth (mm)	361.5	362.5	-0.98	0.326	No	-0.048

Interpretation: Waist circumference and weight are statistically significant ($p < 0.05$), meaning the differences are unlikely to be due to chance. However, all Cohen's d values are below 0.1 — well below the 0.2 threshold for even a "small" effect. This distinction between statistical and practical significance is critical: the large resampled $n=2,000$ gives sufficient statistical power to detect even trivially small differences. The answer to RQ4 is therefore: the differences are statistically real, but not practically large enough to justify region-specific sizing adjustments based on body measurements alone.

7. Impact and Limitations

Potential Implications

This analysis contributes to two conversations. First, at the regional level, it provides data-driven evidence that Washington state's female population is demographically distinct from the national average in ways that directionally support carrying more petite-range inventory at WA locations. Second, and more broadly, the finding that over 50% of American women fall outside Ann Taylor's published size range suggests that the industry's sizing standards are misaligned with the general population — a finding with implications for all U.S. apparel retailers, not just Ann Taylor. The comparison with Korean retailers' more granular size guides suggests that publishing additional dimensions could meaningfully reduce return rates.

Limitations

Population mismatch: NHANES represents all civilian U.S. women aged 20+, which is a far broader group than Ann Taylor's actual customer base. The typical Ann Taylor shopper is likely younger, slimmer, and more urban than the NHANES average, which inflates the "Above Range" proportion in the RQ2 analysis.

ANSUR II sample bias: ANSUR II is a military sample — younger, fitter, and more physically homogeneous than the civilian population. The proxy filtering approach partially mitigates this, but the structural measurement estimates remain subject to this systematic bias.

ANSUR II race/ethnicity limitation: ANSUR II contains no race/ethnicity variable, preventing demographic weighting for structural measurements. Structural estimates are therefore based on body size profiles rather than demographic composition, which is a known methodological

limitation acknowledged throughout the analysis.

Dataset age: ANSUR II (2012) and NHANES (2015-2018) are 8-13 years old. Body measurement trends — particularly weight-related measurements — may have shifted since collection. The structural measurement estimates are likely more stable over time than circumference estimates.

Waist-only size assignment: The RQ2 size distribution analysis uses waist circumference as the sole sizing variable. In practice, Ann Taylor's size guide uses bust, waist, and hip together. Using a single dimension produces a simplified but not fully representative alignment.

Ann Taylor sizing stasis: Ann Taylor's published size guide may not have been updated recently. Vanity sizing trends — where brands gradually enlarge garments while keeping size labels constant — mean the published guide may lag behind actual product dimensions.

8. Challenge Goals

1. Multiple Datasets

Three primary datasets (ANSUR II, NHANES, ACS PUMS) plus the Ann Taylor size guide were combined to answer the research questions. The core join operation maps NHANES RIDRETH3 race/ethnicity codes to ACS PUMS RAC1P/HISP labels, then applies PUMS-derived demographic weights to NHANES group means — producing Washington-representative body measurement estimates. This join is central to RQ1, RQ2, and RQ4. One deviation from the proposal: ANSUR II was excluded from the demographic weighting join due to the absence of a race/ethnicity variable, requiring the proxy filtering approach instead.

2. Result Validity

RQ4 applied independent samples t-tests (`scipy.stats.ttest_ind`) with a significance threshold of $\alpha = 0.05$, and computed Cohen's d effect sizes to assess practical significance alongside statistical significance. The distinction between the two — statistically significant but negligible effect sizes — is a key finding of the project and demonstrates that result validity testing can produce nuanced, non-obvious conclusions. Test assumptions were verified during the EDA phase: normality was confirmed via distribution plots, and large sample sizes justified t-test use despite slight skewness in waist circumference.

9. Work Plan Evaluation

The original work plan estimated 27 total hours across 6 tasks. Several tasks deviated significantly from estimates:

Task	Original	Actual	Reason for Deviation
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1. Data acquisition	3 hrs	3 hrs	On track; national PUMS split into 4 parts was unexpected but minor
2. Cleaning + preprocessing	5 hrs	8 hrs	ANSUR II race/ethnicity absence required full methodology redesign; Ann Taylor size guide expanded from 1 to 4 DataFrames
3. RQ1 + RQ3 analysis	6 hrs	7 hrs	ANSUR II proxy filtering approach added significant complexity
4. RQ2 Ann Taylor alignment	4 hrs	5 hrs	Distribution-based approach more complex than mean comparison; "Above Range" category required additional debugging
5. RQ4 statistical testing	5 hrs	4 hrs	Resampling approach was straightforward once weighted std was ready
6. Report + code cleanup	4 hrs	6 hrs	More sections and visualizations than originally estimated

The largest deviation was in Task 2, where the discovery that ANSUR II lacks race/ethnicity data required a significant rethinking of the entire analysis approach. In hindsight, verifying dataset schemas before finalizing the proposal methodology would have prevented this surprise.

10. Testing

A separate `test_eda.py` module contains 6 unit tests verifying core analysis functions in `analysis.py` using small synthetic DataFrames with known expected outputs. All 6 tests pass. Tests were designed to be independent of the full datasets — using hand-constructed DataFrames with computable expected values — so that correctness can be verified without downloading any data files.

Test	What It Verifies
<code>test_summarize_dataset</code>	Seven-number summary values (mean height=165.0, waist=67.5) and categorical counts on a 4-row synthetic DataFrame
<code>test_compute_demographic_weights</code>	Weights sum to 1.0; Hispanic proportion = $150/500 = 0.30$; Hispanic and Asian labels present
<code>test_compute_weighted_means</code>	Weighted mean with equal 50/50 group weights: waist=700.0mm, height=1600.0mm
<code>test_load_filters_females_only</code>	<code>SEX == 2</code> filter produces 3 female rows from a 5-row test DataFrame
<code>test_missing_data_detection</code>	<code>isnull().sum()</code> correctly identifies 1 missing value per column and 2 total
<code>test_loadanntaylor</code>	Four DataFrames returned with correct row counts (11, 11, 10, 11); inch-to-cm conversion: $24.5 \text{ in} \times 2.54 = 62.23 \text{ cm}$

11. Citations and Collaboration

Datasets

- ANSUR II (Women): Civilian American Anthropometric Survey, 2012. Penn State OPEN Design Lab. <https://www.openlab.psu.edu/ansur2/>
- NHANES 2015-2018 (Women): National Health and Nutrition Examination Survey, Penn State preprocessed version. <https://www.openlab.psu.edu/nhanes/>
- ACS PUMS 2020-2024: American Community Survey 5-Year Public Use Microdata Sample. U.S. Census Bureau. <https://www.census.gov/programs-surveys/acs/microdata/access.html>
- Ann Taylor Size Guide: Manually transcribed from size chart modal at [anntaylor.com](https://www.ann-taylor.com), May 2026.

Tools and Libraries

- Python 3.13, pandas, matplotlib, seaborn, scipy — standard data science stack.
- ReportLab — used for programmatic PDF report generation.

AI Assistance

Claude (Anthropic) was used as an AI assistant throughout this project to help structure research questions, draft report sections, review Python code for correctness and CSE 163 style compliance, explain unfamiliar functions and statistical concepts, and debug code errors. All datasets were independently downloaded. All code was written, reviewed, and understood by the author before submission. All analytical decisions and interpretations are the author's own.